Algebraic K-theory
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Preface

There will be some gaps in explanation, either due to the lecturer's admission or my own lack of understanding. In particular, many "proofs" are sketches of proofs. Gaps due to my own misunderstanding will be indicated by three red question marks: ???. More generally, my own questions about the material will also be in red. Things like "**Question**" will be questions posed by the lecturer. Feel free to reach out to me with explanations.

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Objects of study:

- Algebraic K-groups, i.e. for ring R, groups $K_n(R)$ for $n \geq 0$.
- Steinberg module, a representation $St_n(F)$ of $GL_n(F)$ for F a field.

We assume throughout the course that rings are commutative. R will always denote a ring and F will always denote a field.

Definition 1.1. An R-module P is projective if there is an R-module P' such that $P \oplus P'$ is free.

Definition 1.2 (1957, Grothendieck)**.** The group $K_0(R)$ is generated by isomorphism classes of finitely generated projective R-modules, subject to the relation $[P \oplus Q] = [P] + [Q]$.

Example 1.1. If R is a PID (including a field), then $K_0(R) \cong \mathbb{Z}$, since every finitely generated projective R-module is free.

Example 1.2. If F is a number field with integer ring $\mathcal{O}_F$, then $K_0(\mathcal{O}_F) \cong \mathbb{Z} \oplus Cl(F)$, where $Cl(F)$ is the ideal class group.

Definition 1.3 (1964, Bass)**.** Consider embeddings $GL_n(R) \hookrightarrow GL_{n+1}(R)$ where A is sent to the block matrix $\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$. Let $GL(R) = \varinjlim GL_n(R)$. Then $K_1(R)$ is the abelianization of $GL(R)$, i.e. $GL(R)/[GL(R), GL(R)]$.

Another description of $[GL(R), GL(R)]$ is given as follows. For $i \neq j$ in $\{1, \ldots, n\}$, let $e_{ij}(a) = \mathrm{id} + a\delta_{ij}$ be the matrix in $GL_n(R)$ with 1 on the diagonal, a in the (i, j) entry, and 0 elsewhere. This is called an elementary matrix. Let $E_n(R)$ be the subgroup of $GL_n(R)$ generated by the elementary matrices. Then let $E(R) = \varinjlim E_n(R)$.

Theorem 1.1 (Whitehead)**.** $E(R) = [GL(R), GL(R)]$.

Thus $K_1(R) = GL(R)/E(R)$.

Example 1.3. If F is a field, then $K_1(F) \cong F^\times$. We also have $K_1(\mathbb{Z}) \cong \mathbb{Z}/2$.

Remark. If R is a regular Noetherian ring, then $K_i(R[t]) \cong K_i(R)$ for $i = 0, 1$.

Elementary matrices have certain "universal" relations that do not depend on the ring R, which come in three forms. The first is $e_{ij}(a)e_{ij}(b) = e_{ij}(a + b)$. This can be depicted with a picture:

$$\begin{tikzcd}
 & \bullet \arrow[dr, "e_{ij}(b)"] & \\
\bullet \arrow[ur, "e_{ij}(a)"] \arrow[rr, "e_{ij}(a+b)"'] & & \bullet
\end{tikzcd}$$

If $i \neq l$ and $j \neq k$, then $[e_{ij}(a), e_{kl}(b)] = \mathrm{id}$. This is depicted by a square. If $i \neq k$, then $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$, which is depicted by a pentagon. Note: lecturer did not label the edges of the square and pentagon, so I will not include their pictures. check the relations!

The Steinberg group (unrelated to Steinberg module) $St_n(R)$ has generators $x_{ij}(a)$ for $i \neq j$ in $\{1, \ldots, n\}$ and relations given by the "universal" relations of elementary matrices. In particular, $x_{ij}(0)x_{ij}(0) = x_{ij}(0)$ shows that $x_{ij}(0)$ is the identity of $St_n(R)$ for all $i \neq j < n$. We can take the colimit $St(R)$, and it surjects onto $E(R)$. The kernel of the surjection is the center $Z(St(R))$. The corresponding short exact sequence is called the universal central extension of $E(R)$.

Definition 1.4 (1967, Milnor)**.** $K_2(R) = Z(St(R))$.

Example 1.4 (Matsumoto)**.** For a field F, $K_2(F) \cong F^\times \otimes F^\times / \langle x \otimes (1-x) \rangle$.

Remark. $K_2(R) \cong H_2(E(R), \mathbb{Z})$.

Example 1.5. $K_2(\mathbb{Z}) \cong \mathbb{Z}/2$.

Remark. $K_3(R) \cong H_3(St(R), \mathbb{Z})$. For F a field, $K_3(F) \otimes \mathbb{Q}$ is understood, by a result of Suslin.

Throughout 1968-1970, many definitions of $K_n(R)$ appeared. They are equivalent for "good" rings. Quillen's definition won out.

Definition 1.5 (Quillen)**.** Let $Pr(R)$ be the category of finitely generated projective modules. Let $QPr(R)$ be the category of finitely generated modules, with a morphism between P_1 and P_2 being a "roof diagram" (with $Q \in Pr(R)$).

$$\begin{tikzcd} & Q \arrow[dl, two heads] \arrow[dr, hook] & \\ P_1 & & P_2 \end{tikzcd}$$

We define $K_n(R) = \pi_{n+1}(B[QPr(R)])$, where $B[QPr(R)]$ is the classifying space of $QPr(R)$, i.e. a topological space built out of simplices corresponding to chains of morphism in $QPr(R)$.

Definition 1.6. For F a field, we define the Steinberg module $St_n(F)$ to have generators given by n-tuples of vectors in F^n, i.e. $[v_1, \ldots, v_n] \in F^n$. The relations are given as follows:

- If $\lambda \in F^\times$, then $[\lambda v_1, v_2, \ldots, v_n] = [v_1, \ldots, v_n]$.
- If $\sigma \in S_n$, then $[v_{\sigma(1)}, \ldots, v_{\sigma(n)}] = (-1)^\sigma [v_1, \ldots, v_n]$.
- $\sum (-1)^i [v_1, \ldots, \widehat{v_i}, \ldots, v_{n+1}] = 0$.
- If $v_1, \ldots, v_n$ are linearly dependent, then $[v_1, \ldots, v_n] = 0$.

It has a natural linear $GL_n(F)$ action.

Quillen constructed a spectral sequence $H_\bullet(GL_\bullet(F), St_\bullet(F)_{\mathbb{Q}}) \Rightarrow H_\bullet(B[QPr(R)]; \mathbb{Q})$.

Theorem 1.2. *$H_i(GL_n(F), St_n(F)_{\mathbb{Q}}) = 0$ for $n \geq 2$ and $i \geq n-2$.*

The group $H_{n-1}(GL_n(F), St_n(F)_{\mathbb{Q}})$ is very rich and is related to scissors congruence, polylogarithms, hyperbolic geometry...

There is another spectral sequence $H_\bullet(GL_\bullet(F), St^\infty_\bullet(F)_{\mathbb{Q}}) \Rightarrow K_0(F)_{\mathbb{Q}}$, where St^∞ is the stable Steinberg module.

2 Mar 25

We give another definition of the Steinberg module $St_n(F)$. The Tits building of F^n, J_{F^n}, is a simplicial complex where p-simplicies are partial p-flags inside F^n. A partial p-flag is a strictly increasing chain of $p+1$ nonzero proper subspaces of F^n. For $n=2$ there are only 0-simplices, which correspond to lines in F^2. For $n=3$, the 0-simplices are lines and planes in F^3, and the 1-simplices are between a line and a plane whenever the plane contains the line.

Theorem 2.1 (Solomon-Tits)**.** *J_{F^n} is homotopy equivalent to a bouquet of spheres S^{n-2}.*

The Steinberg module is $St_n(F) = \widetilde{H}_{n-2}(J_{F^n})$.

The highest dimensional chains $C_{n-2}(J_{F^n})$ are (linear combinations of) complete flags. Given a basis $v_1, \dots, v_n$ in F^n, one can associate a chain via $\sum_{\sigma \in S^n} (-1)^\sigma [\langle v_{\sigma(1)} \rangle, \langle v_{\sigma(1)}, v_{\sigma(2)} \rangle, \dots]$, and it is in fact a closed chain, hence giving an element in $St_n(F)$. This corresponds to the element $[v_1, \dots, v_n]$ from our previous definition of $St_n(F)$.

Last time we mentioned the following result:

Theorem 2.2. *$H_i(GL_n(F), St_n(F)_{\mathbb{Q}}) = 0$ for $n \geq 2$ and $i \geq n-2$.*

There is a related conjecture:

Conjecture 2.1 (Church-Farb-Pitman)**.** *$H_i(GL_n(\mathbb{Z}), St_n(\mathbb{Q})) = 0$ for $n \geq 2$ and $i \geq n-2$.*

The conjecture (should) be true for $n \leq 4$, as well as for $i = 0, 1, 2$. We will do $i = 0, 1$ later on.

Let G be a group and M a G-module. $H_0(G, M)$ is the coinvariants of M, denoted $M_G = M/(m - gm \mid m \in M, g \in G)$.

Theorem 2.3 (Ash-Rudolph)**.** *$St_n(\mathbb{Q})$ is generated as an abelian group by integral "apartments", i.e. tuples $[v_1, \dots, v_n]$ where $v_i \in \mathbb{Z}^n$ and $\det(v_1, \dots, v_n) = \pm 1$.*

Corollary 2.1. *$H_0(GL_n(\mathbb{Z}), St_n(\mathbb{Q})) = 0$ for $n \geq 2$.*

Theorem 2.4 (Bykowski)**.** *$St_n(\mathbb{Q})$ is spanned as a vector space by integral apartments subject to the following relations (for integral apartments only):*

- $[-v_1, v_2, \dots, v_n] = [v_1, \dots, v_n]$.
- *If $\sigma \in S_n$, then $[v_{\sigma(1)}, \dots, v_{\sigma(n)}] = (-1)^\sigma [v_1, \dots, v_n]$.*
- $[v_1, v_2, v_3, \dots] + [v_2, v_1 + v_2, v_3, \dots] + [v_1 + v_2, v_1, v_3, \dots] = 0$.

The last relation is similar to the relations appearing in Milnor K-theory:

Definition 2.1 (Milnor K-theory)**.** $K^M(F) = \bigoplus (F^\times)^{\otimes n}/(x \otimes (1-x))$. If one is "relaxed about 2-torsion", then $K_n^M(F) = \bigwedge^n F^\times/(x \wedge (1-x) \wedge y_3 \wedge \cdots \wedge y_n)$.

Remark. $\frac{a}{b} \wedge (1 - \frac{a}{b}) = a \wedge (b-a) - b \wedge (b-a) - a \wedge b$. (???)

2.1 End of overview

We return to discuss $K_1(R)$. Recall the elementary matrices and their subgroup $E_n(R)$ in $GL_n(R)$.

Theorem 2.5 (Suslin)**.** *$E_n(R)$ is normal in $GL_n(R)$ for R commutative and $n \geq 3$.*

Note. We assume R is commutative throughout the course even though much of the theory works with R noncommutative. However, the above theorem does require R to be commutative.

add stuff on unimodular and elementary matrices

Theorem 2.6 (Whitehead)**.** $E(R) = [GL(R), GL(R)]$.

Proof. First we show $E \subset [GL, GL]$ (we drop the R dependence notation). We use the following relation: If $i \neq k$, then $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$. Then $e_{ij}(a) = [e_{ik}(a), e_{kj}(1)]$ for any k different from i, j (we can pick such an index since we are working with arbitrarily large matrices). We claim that if $A, B \in GL_n$, then $ABA^{-1}B^{-1} \in E_{2n}$. We have

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The above relation is valid even if we take $a \in GL_n$. In particular, taking $a = \mathrm{id}_n$, we get $\begin{pmatrix} 0 & -\mathrm{id}_n \\ \mathrm{id}_n & 0 \end{pmatrix} \in E_{2n}$. It follows that $\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} \in E_{2n}$. Then

$$\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} \begin{pmatrix} B & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} (BA)^{-1} & 0 \\ 0 & BA \end{pmatrix} = \begin{pmatrix} ABA^{-1}B^{-1} & 0 \\ 0 & \mathrm{id}_n \end{pmatrix} \in E_{2n}.$$

We have $K_1(R) = GL(R)/E(R) = GL(R)/[GL(R), GL(R)] = H_1(GL(R), \mathbb{Z})$. We have $K_1(F) = F^\times$ and $K_1(\mathbb{Z}) = \mathbb{Z}^\times = \{\pm 1\}$.

Recall the Steinberg group defined last time. $St_n(R)$ comes with a surjection to $E_n(R)$, inducing a surjection from $\phi : St(R) \to E(R)$.

Theorem 2.7. $\ker(\phi) = Z(St(R))$.

Proof. ϕ must send $Z(St_n(R))$ into $Z(E_n(R))$. Any matrix in $Z(E_n(R))$ is diagonal, with all diagonal entries equal to the same element of $R^\times$. Upon passing to the direct limit, we see that the only element of $Z(E(R))$ is the identity. Thus

$Z(St(R)) \subset \ker(\phi)$.

Now let $y \in \ker \phi$. There is some N such that y is a product of generators $x_{ij}(a)$ with all i, j less than N. Take $n > N$ and denote by P_n the group generated by elements of the form $x_{in}(a)$ for $i < n$. Then any element of P_n can be uniquely expressed in the form $x_{1n}(a_1) \cdots x_{n-1,n}(a_{n-1})$. presumably that follows from playing with relations, check! It follows that ϕ maps P_n isomorphically to its image, which consists of matrices of the form

$$\begin{pmatrix} 1 & 0 & \cdots & a_1 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1} \\ 0 & \cdots & 0 & 1 \end{pmatrix}$$

One can show that if $i, j < n$, then $x_{ij}(a) P_n x_{ij}(a)^{-1} \subset P_n$. check it! Thus $y P_n y^{-1} \subset P_n$. So, for any $p \in P_n$, we have $ypy^{-1} \in P_n$. But $\phi(ypy^{-1}) = \phi(p)$ since $y \in \ker \phi$, and ϕ is injective on P_n, so $ypy^{-1} = p$. It follows that y commutes with $x_{in}(a)$ for all $i < n$. A similar setup shows that y commutes with $x_{nj}(a)$ for all $j < n$. Thus, y commutes with $x_{ij}(a) = [x_{in}(a), x_{nj}(1)]$ for all $i, j < n$. Since n was arbitrary, it follows that $y \in Z(St(R))$.

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Let ϕ be the natural map $St(R) \to E(R)$. Let $A, B \in E(R)$ commute. Then for $a \in \phi^{-1}(A)$ and $b \in \phi^{-1}(B)$, the element $[a, b]$ depends only on A and B, and furthermore lands in $\ker(\phi) = K_2(R)$. We denote it by $A * B$.

Properties of $*$ check!:

- $B * A = (A * B)^{-1}$.
- $(A_1 A_2) * B = (A_1 * B)(A_2 * B)$.
- $(PAP^{-1}) * (PBP^{-1}) = A * B$.

For $u, v \in R^\times$, we define

$$\{u, v\} = \begin{pmatrix} u & 0 & 0 \\ 0 & u^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} v & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & v^{-1} \end{pmatrix}.$$

For $u, v, w \in R^\times$, we have $\{uv, w\} = \{u, w\}\{v, w\}$ and $\{u, vw\} = \{u, v\}\{v, w\}$. should follow from properties of *, but check!

Theorem 3.1. *Let $u, 1-u \in R^\times$. Then $\{u, 1-u\} = 1$ in $K_2(R)$.*

Proof. We fix $n \geq 3$ and drop R from our notation. Throughout the proof, the letters u and v will always refer to units in R, and u is not always assumed to be a unit such that $1-u$ is also a unit.

We begin by introducing the following elements of St_n, for $1 \leq i, j \leq n$ and $i \neq j$:

$$w_{ij}(u) = x_{ij}(u) x_{ji}(-u^{-1}) x_{ij}(u).$$

As a matrix it is $\mathrm{id}_n - \delta_{ii} - \delta_{jj} + u\delta_{ij} - u^{-1}\delta_{ji}$, i.e. it has u in the i, j entry, $-u^{-1}$ in the j, i entry, 1 in the k, k entry for $k \neq i, j$, and 0 elsewhere. The $w_{ij}(u)$ generate a group that we call W_n. The image $\phi(W_n)$ is the monomial matrices. Now, we let $h_{ij}(u) = w_{ij}(u) w_{ij}(-1)$. As a matrix it is $\mathrm{id}_n - \delta_{ii} - \delta_{jj} + u\delta_{ii} + u^{-1}\delta_{jj}$, i.e. it has u in the i, i entry, u^{-1} in the j, j entry, 1 in the k, k entry for $k \neq i, j$, and 0 elsewhere. The $h_{ij}(u)$ generate a group H_n, which is contained in W_n since each $h_{ij}(u)$ is in W_n.

Exercise: $w_{ij}(u)^{-1} = w_{ij}(-u) = w_{ji}(u^{-1})$. check!

Lemma 3.1 (Key Lemma)**.** *Conjugation by an element of W_n sends an element of the form $x_{ij}(a)$ to an element of the form $x_{kl}(b)$, and it sends something of the form $w_{ij}(u)$ to an element of the form $w_{k'l'}(v)$.*

Remark. Lemma 3.1 is very powerful and essentially encodes all of the Steinberg relations we will use. We can use it to compute conjugation in the Steinberg group by computing them in matrices.

Let $C_n = W_n \cap \ker(\phi)$.

Corollary 3.1. $C_n \subset Z(St_n)$.

Proof of Corollary. Let $w \in C_n$. Then we have $\phi(wx_{ij}(a)w^{-1}) = \phi(x_{ij}(a))$. From Lemma 3.1, we can write $wx_{ij}(a)w^{-1} = x_{kl}(b)$, so we get $e_{kl}(b) = e_{ij}(a)$. Thus $i = k, j = l, a = b$, so $wx_{ij}(a)w^{-1} = x_{ij}(a)$. Thus $w \in Z(St_n)$.

Lemma 3.2. $\{u, v\} = [h_{12}(u), h_{13}(v)] = h_{13}(uv)h_{13}(u)^{-1}h_{13}(v)^{-1}$.

Proof of Lemma 3.2. The first equality holds since $h_{12}(u)$ and $h_{13}(v)$ represent the matrices involved in the definition of $\{u, v\}$. We can rewrite

$$\begin{aligned} h_{12}(u)h_{13}(v)h_{12}(u)^{-1} &= h_{12}(u)w_{13}(v)w_{13}(-1)h_{12}(u)^{-1} \\ &= [h_{12}(u)w_{13}(v)h_{12}(u)^{-1}][h_{12}(u)w_{13}(-1)h_{12}(u)^{-1}]. \end{aligned}$$

Since $h_{12}(u) \in W_n$, Lemma 3.1 says we can compute both of the bracketed terms above by doing the computation in matrices. We find

$$\begin{aligned} &\begin{pmatrix} u & 0 & 0 \\ 0 & u^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & v \\ 0 & 1 & 0 \\ -v^{-1} & 0 & 0 \end{pmatrix} \begin{pmatrix} u^{-1} & 0 & 0 \\ 0 & u & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ = &\begin{pmatrix} 0 & 0 & uv \\ 0 & u^{-1} & 0 \\ -v^{-1} & 0 & 0 \end{pmatrix} \begin{pmatrix} u^{-1} & 0 & 0 \\ 0 & u & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & uv \\ 0 & 1 & 0 \\ -v^{-1}u^{-1} & 0 & 0 \end{pmatrix}, \end{aligned}$$

so that $h_{12}(u)w_{13}(v)h_{12}(u)^{-1} = w_{13}(uv)$. Taking $v = -1$ we get the other bracketed term is $w_{13}(-u)$, which by the exercise is $w_{13}(u)^{-1}$. Thus,

$$\begin{aligned} h_{12}(u)h_{13}(v)h_{12}(u)^{-1} &= w_{13}(uv)w_{13}(u)^{-1} \\ = [w_{13}(uv)w_{13}(-1)][w_{13}(u)w_{13}(-1)]^{-1} &= h_{13}(uv)h_{13}(u)^{-1}. \end{aligned}$$

Multiplying both sides by $h_{13}(v)^{-1}$ gives the desired result.

Finally, let u, v be units with $u + v = 1$. By Lemma 3.2, we want to show $h_{13}(uv) = h_{13}(u)h_{13}(v)$. Symbolically it is enough to show $h_{12}(uv) = h_{12}(u)h_{12}(v)$. By expanding the definition of h and canceling a factor of $w_{12}(-1)$, we are reduced to showing $w_{12}(uv) = w_{12}(u)w_{12}(-1)w_{12}(v)$. By the exercise, we can rewrite $w_{12}(-1) = w_{21}(1)$, which by definition is $x_{21}(1)x_{12}(-1)x_{21}(1)$. Using Lemma 3.1, we can compute $w_{12}(u)x_{21}(1)w_{12}(u)^{-1} = x_{12}(-u^2)$ in matrices:

$$\begin{pmatrix} 0 & u \\ -u^{-1} & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & -u \\ u^{-1} & 0 \end{pmatrix} = \begin{pmatrix} u & u \\ -u^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -u \\ u^{-1} & 0 \end{pmatrix} = \begin{pmatrix} 1 & -u^2 \\ 0 & 1 \end{pmatrix}.$$

Similarly, we can compute $w_{12}(v)^{-1}x_{21}(1)w_{12}(v) = x_{12}(-v^2)$. It follows that

$$\begin{aligned} w_{12}(u)w_{12}(-1)w_{12}(v) &= w_{12}(u)x_{21}(1)x_{12}(-1)x_{21}(1)w_{12}(v) \\ &= x_{12}(-u^2)w_{12}(u)x_{12}(-1)w_{12}(v)x_{12}(-v^2). \end{aligned}$$

Next, we expand the definition of $w_{12}(u)$ and $w_{12}(v)$, and then use Steinberg relations and $u+v=1$ to consolidate:

$$\begin{aligned}
w_{12}(u)w_{12}(-1)w_{12}(v) &= x_{12}(-u^2)w_{12}(u)x_{12}(-1)w_{12}(v)x_{12}(-v^2) \\
= x_{12}(-u^2)x_{12}(u)x_{21}(-u^{-1})x_{12}(u)x_{12}(-1)x_{12}(v)x_{21}(-v^{-1})x_{12}(v)x_{12}(-v^2) \\
&= x_{12}(u-u^2)x_{21}(-u^{-1})x_{12}(u-1+v)x_{21}(-v^{-1})x_{12}(v-v^2) \\
&= x_{12}(uv)x_{21}(-u^{-1})x_{21}(-v^{-1})x_{12}(uv) \\
&= x_{12}(uv)x_{21}(-u^{-1}-v^{-1})x_{12}(uv) \\
&= x_{12}(uv)x_{21}(-u^{-1}v^{-1})x_{12}(uv) = w_{12}(uv).
\end{aligned}$$

Thus, we are done.

Remark. Recall Milnor K_2 is $K_2^M(F) = F^\times \otimes F^\times/(u \otimes (1-u))$. The theorem implies there is a map $K_2^M(F) \to K_2(F)$. It is a theorem of Matsumoto that this map is an isomorphism. A similar result holds for local rings.

We now do some work with $K_2^M(F)$. We use additive notation for this group, and we use $\{x,y\}$ to denote the image of $x \otimes y$ in $K_2^M(F)$. In particular, $\{x, 1-x\} = 0$. Some observations:

- $\{x,-x\} = 0$. Indeed, using $1-x = -x(1-x^{-1})$ we have

$$\{x,-x\} = \{x,1-x\} - \{x,1-x^{-1}\} = 0 + \{x^{-1},1-x^{-1}\} = 0.$$

- $\{x,y\} = -\{y,x\}$. This follows from:

$$\begin{aligned}
0 = \{xy,-xy\} &= \{x,-xy\} + \{y,-xy\} \\
= \{x,-x\} + \{x,y\} + \{y,x\} + \{y,-y\} &= \{x,y\} + \{y,x\}.
\end{aligned}$$

  As a consequence, we also have $2\{x,x\} = 0$.

Theorem 3.2. $K_2^M(\mathbb{F}_q) = 0$.

Proof. Let x generate $\mathbb{F}_q^\times$. Then it suffices to show $\{x,x\} = 0$. If q is even, then $x = -x$, so this follows from $\{x,-x\} = 0$. If q is odd, first note that $\{x^m, x^n\} = mn\{x,x\} = \{x,x\}$ if m,n are both odd, since $2\{x,x\} = 0$. Then it is a general fact that there are nonsquare x_1, x_2 such that $x_1 + x_2 = 1$, so that $\{x_1, x_2\} = 0$. Thus $\{x,x\} = 0$, as desired.

Definition 3.1. A map from $K_2^M(F)$ to an abelian group A is called a **symbol**.

To give a symbol $c : K_2^M(F) \to A$, it is equivalent to assign elements $c(x,y) \in A$ for all units $x, y \in F$, such that $c(xy,z) = c(x,z) + c(y,z)$ and $c(x,1-x) = 0$.

Example 3.1. There is a symbol $K_2^M(\mathbb{R}) \to \mathbb{Z}/2$ given by $c(x,y) \neq 0$ if and only if $x, y < 0$.

Example 3.2 (Tame symbol)**.** Let F be a field with a discrete valuation ν. Let $\mathcal{O}_F$ be the valuation ring, $\mathfrak{m}_F$ its maximal ideal, and $\overline{F}$ the residue field. Let $\pi \in F^\times$ have $\nu(\pi) = 1$. Then there is a map $\mathrm{res} : K_2^M(F) \to \overline{F}^\times$ satisfying

$$\mathrm{res}\{x, y\} = (-1)^{\nu(x)\nu(y)}\overline{y^{\nu(x)}x^{-\nu(y)}},$$

where the overline means image in $\overline{F}$. In terms of π and units $u, v \in \mathcal{O}_F^\times$, we have

$$\mathrm{res}\{\pi^n u, \pi^m v\} = (-1)^{nm}\overline{v^n u^{-m}}.$$

This map is characterized by the fact that $\mathrm{res}\{u, v\} = 0$ and $\mathrm{res}\{\pi, u\} = \overline{u}$.

Exercise: show $\mathrm{res}\{x, 1 - x\} = 0$.

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We continue discussing symbols.

Example 4.1 (Differential symbol)**.** Let F be characteristic 0. Let $\Omega^1_{F/\mathbb{Q}}$ be the space of differentials, which is an F-vector space with generators dx for $x \in F$ such that $d(x+y) = dx + dy$, $d(xy) = xdy + ydx$, and $d(a) = 0$ for $a \in \mathbb{Q}$. We also let $\Omega^n_{F/\mathbb{Q}} = \bigwedge^n \Omega^1_{F/\mathbb{Q}}$. There is a differential symbol $K_2^M(F) \to \Omega^2_{F/\mathbb{Q}}$ given by sending $\{x, y\}$ to $\frac{dx}{x} \wedge \frac{dy}{y}$.

Now, recall the tame symbol (Example 3.2) from last time. Each p-adic valuation on $\mathbb{Q}$ gives a map $\mathrm{res}_p : K_2^M(F) \to \mathbb{F}_p^\times$. We aim to prove the following:

Theorem 4.1 (Tate)**.** *There is a short exact sequence*

$$0 \to \mathbb{Z}/2 \xrightarrow{-1 \mapsto \{-1,-1\}} K_2^M(\mathbb{Q}) \xrightarrow{\bigoplus_p \mathrm{res}_p} \bigoplus_p \mathbb{F}_p^\times \to 0.$$

Proof. We define a filtration $L_\bullet$ on $K_2^M(\mathbb{Q})$ by setting L_n to be spanned by $\{x, y\}$ for $x, y \in \mathbb{Z}$ with $|x|, |y| \leq n$. Then L_1 is one-dimensional and spanned by $\{-1, -1\}$. The quotients L_n/L_{n-1} are only nontrivial when n is prime.

Lemma 4.1. *L_p/L_{p-1} is cyclic of order $p-1$.*

Let us first finish the proof of the theorem assuming the lemma. Clearly res_p vanishes on L_{p-1}. Then one can show that the induced map $\mathrm{res}_p : L_p/L_{p-1} \to \mathbb{F}_p^\times$ is an isomorphism, and the theorem follows.
Now, we prove the lemma.

Proof of Lemma 4.1. We construct a section $s_p : \mathbb{F}_p^\times \to L_p/L_{p-1}$ to res_p by $s_p(x) = \{p, x\}$, where we interpret $x \in \mathbb{F}_p^\times$ as an integer satisfying $1 \leq x \leq p-1$. We need to check that this is a homomorphism.

Remark. The appearance of $\mathbb{Z}/2$ in Theorem 4.1 is due to $K_2(\mathbb{Z}) = \mathbb{Z}/2$.

Exercise. The following sequence is exact:

$$0 \to K_2^M(F) \to K_2^M(F(t)) \to \bigoplus_{p \in \mathbb{A}^1_F} (F[t]/\delta_p(t))^\times \to 0.$$

(not sure what the δ_p is or if that is the right letter)

For p an odd prime, we define the p Hilbert symbol $K_2^M(\mathbb{Q}) \to \mathbb{F}_p^\times/(\mathbb{F}_p^\times)^\square = \{\pm 1\}$ by composing res_p with the p Legendre symbol:

$$((x, y))_p = \left(\frac{\mathrm{res}_p\{x, y\}}{p}\right).$$

We also define

$$((x,y))_\infty = \begin{cases} -1 & x,y<0 \\ 1 & \text{else} \end{cases}$$

and TODO: 2, quad form, exers.

Our next goal is the following theorem:

Theorem 4.2. *$K_2^M(F) \to K_2(F)$ is surjective.*

todo: add recap of proof of theorem from last time

Lemma 4.2. *C_n is generated by $\{u,v\}$ for $u,v \in R^\times$.*

Proof. There is a map $W_n \to S_n$ given by looking at the corresponding permutation of the matrix representation of an element of W_n. All H_n are represented by diagonal matrices, so this descends to a map $W_n/H_n \to S_n$.

Claim 4.1 (Step 1)**.** *The map $W_n/H_n \to S_n$ is an isomorphism.*

Proof of Step 1. First we show that in W_n/H_n, $w_{ij}(u) = w_{ij}(v)$ for all u,v, so we just call it w_{ij}. add. Then w_{ij} goes to the transposition (ij), and S_n is generated by transpositions, so the map is surjective. We can construct an inverse by showing that the assignment $(i,i+1) \mapsto w_{i,i+1}$ is a well-defined homomorphism inverse to ours.

Claim 4.2 (Step 2)**.** *For i,j,k distinct, we have*

$$h_{ij}(u)^{-1}h_{jk}^{-1}(u)h_{ki}(u)^{-1} = 1.$$

Furthermore, $h_{ij}(u)^{-1} = h_{ji}(u)$.

Proof of Step 2. todo

As a result, H_n is generated by $h_{1j}(u)$ (I think?)

Claim 4.3 (Step 3)**.** *For i,j,k distinct, we have*

$$h_{ij}(u)^{-1}h_{jk}^{-1}(u)h_{ki}(u)^{-1} = 1.$$

Furthermore, $h_{ij}(u)^{-1} = h_{ji}(u)$. todo

Proof of Step 3. todo

We also need the following result that we will not prove:

Proposition 4.1 (Bruhat decomposition for $St_n(F)$)**.** *Let T_n be the subgroup of $St_n(F)$ generated by $x_{ij}(a)$ for $i<j$. Then $St_n(F) = T_nW_nT_n$.*

todo: finish proof of thm